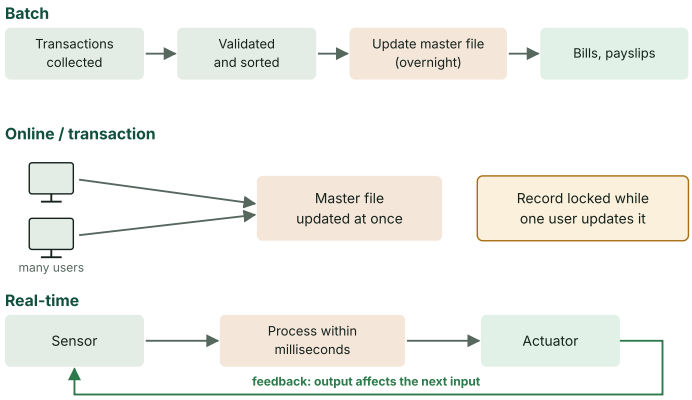
Check digits, batch totals, data capture and processing methods

FD07 FACT FILE

Specification: students should be able to

- Explain the purpose of check digits and calculate and check a Modulus 11 check digit
- Explain the purpose of batch totals, control totals and hash totals and apply them to a batch of documents
- Describe manual and automated methods of data capture
- Describe batch and real-time processing, distinguishing real-time transaction (online) processing from real-time control, and select an appropriate method for a given application
- Explain multitasking, multi-user and multithreading in the context of processing

Key words: check digit, Modulus 11, weighting, remainder, batch total, control total, hash total, data capture, optical mark recognition (OMR), batch processing, master file, transaction file, real-time transaction (online) processing, real-time control, record locking, multitasking, multi-user, multithreading



Batch processing collects transactions and updates the master file later; transaction processing updates it immediately with record locking; real-time processing uses a feedback loop from sensors

Check digits and Modulus 11

Long code numbers such as barcodes, book numbers and account numbers are easy to mistype or misread, so many include a check digit: an extra digit calculated from the others and added to the end. When the number is entered or scanned, the computer repeats the calculation and compares its result with the check digit that was entered. If they disagree, the number is rejected. The Modulus 11 method gives each digit a weighting, starting at 2 for the rightmost digit and rising by one for each position to the left. Each digit is multiplied by its weighting and the products are added to give a checksum. The checksum is divided by 11 and the remainder found. If the remainder is 0 the check digit is 0; if it is 1 the check digit is X, which stands for 10; otherwise the check digit is 11 minus the remainder. Because a check digit can be X, numbers like this are stored as strings. Modulus 11 detects all single-digit errors and all transpositions of two digits, which makes it a powerful validation check, but it only shows that a number is internally consistent, not that it is the correct number.

Calculating and checking a check digit

Take the base number 3604. With four digits the weightings are 5, 4, 3 and 2. Multiplying gives 3 × 5 = 15, 6 × 4 = 24, 0 × 3 = 0 and 4 × 2 = 8, so the checksum is 47. Dividing 47 by 11 gives 4 remainder 3, and 11 − 3 = 8, so the full number is 36048. To check a number that already includes its check digit, give the check digit a weighting of 1 and repeat the calculation: 15 + 24 + 0 + 8 + 8 = 55, and 55 divided by 11 leaves remainder 0, so the number is accepted. If a clerk swaps the first two digits and types 63048, the products become 30, 12, 0, 8 and 8, totalling 58, which leaves a remainder of 3, so the error is detected. When the remainder is 1 the check digit is X; for example 1205 gives a checksum of 23, remainder 1, so the full number is 1205X, and when checking, X counts as 10. Always set the working out in a table of weightings, digits and products.

Batch, control and hash totals

When many paper documents, such as fee forms or timesheets, are typed in as a batch, normal validation of each field cannot show that every document was entered exactly once. Batch totals solve this. A batch total is simply a count of the documents, made by hand before entry and compared with the computer's count afterwards. A control total is the sum of a meaningful field on every document, such as the total fees paid. A hash total is the sum of a field that has no meaning when added up, such as member ID numbers. In each case the manual total is compared with the total calculated by the computer, and any difference shows a mistake. A batch total cannot detect one document missed and another entered twice, because the count is unchanged. A control total may also fail if the two documents happen to have the same value in that field. A hash total of unique IDs is much harder to fool, and when exactly one document is missing, the difference between the totals is that document's ID. The chosen field must vary between documents.

Data capture

Data capture is the process of getting data into a computer system. Manual methods, such as keying data from paper forms, are cheap and flexible but slow and prone to transcription and transposition errors, so validation, verification and batch totals are essential. Online forms capture data at source, so there is no re-keying, and validate each field as it is entered, but users need a device and an internet connection. Automated methods are faster and more accurate for high volumes. Optical mark recognition reads marks in fixed positions on forms such as multiple-choice answer sheets. Barcode and QR code scanners read product and parcel codes, and the check digit confirms the scan was correct. Chip and contactless card readers capture account details without typing. Sensors and data loggers record readings such as temperature or water level continuously and unattended. Automated methods need specialist equipment, and each only works when the data is in the expected form, so a choice of method depends on volume, accuracy needed, cost and the kind of data being captured.

Batch processing

In batch processing, transactions are collected over a period into a transaction file and then processed together at a later time, often overnight, with no user interaction once the job has started. The transaction file is validated, using checks such as batch and control totals, and sorted into the same key order as the master file. The program then reads both files in sequence and produces an updated master file along with outputs such as bills, payslips or re-order lists. Typical applications are payroll, utility and credit card billing, end-of-day stock re-ordering and the overnight processing of bank transactions. Batch processing makes efficient use of computer time because the same operation is repeated on many records, and it can be scheduled when the system is otherwise quiet. Its main drawback is that the master file is only up to date immediately after each run, so it is unsuitable where people need current information, such as available seats. Errors in the data may also only be discovered after the run has finished.

Real-time transaction and control processing

In real-time processing each transaction or input is dealt with as soon as it happens, so the master file is always up to date. It takes two forms. Real-time transaction processing, often called online processing, handles each transaction while the user waits a few seconds; it is used for bookings, cash machines, online shopping and anything where the next user must see the effect of the previous transaction, such as a seat being sold or a password being changed. Because many users may try to change the same record at the same moment, these systems use record locking: while one user's transaction is updating a record, other users cannot change it, which prevents lost updates and double-booking. They need reliable networks and servers that are available at all times. Real-time control goes further: sensor input must be processed within a guaranteed, very short time, so that the output can influence the next input. Anti-lock brakes, traffic lights that respond to queues and heart monitors are examples. They use feedback and must be extremely reliable, often with duplicated hardware, because a late response can be dangerous.

Multitasking, multi-user and multithreading

Modern systems process many jobs at once, and the operating system manages how they share the hardware. Multitasking allows one user to run several programs apparently at the same time: the processor switches rapidly between processes, giving each a short time slice, so that music can play while a document is edited. A multi-user system lets many users share the same computer's resources at the same time, each with their own session, files and permissions; a school file server or a bank's central system are examples, and the operating system must keep users' data separate and secure. Multithreading splits a single program into threads, smaller units of execution that share the program's memory and can run concurrently, so a web browser can download images in one thread while responding to scrolling in another. On a multicore processor threads can genuinely run in parallel, improving performance. In transaction processing systems these ideas combine: many users, each served by threads, update shared records, which is why record locking is required.

T Key terms from the mark schemes

Check digit
An extra digit calculated from the other digits of a code number and added to the end, so the computer can detect entry or reading errors by recalculating it.

Batch total
A validation check in which the number of documents in a batch is counted by hand and compared with the number entered by the computer.

Hash total
A batch total formed by adding up a field whose total has no meaning, such as ID numbers, and comparing it with the computer total.

Real-time transaction processing
Real-time processing in which each transaction is processed as it occurs while the user waits, so the master file is always up to date, e.g. bookings and cash machines.
`Credited: Online (transaction) processing`

Record locking
Preventing other users from changing a record while one user is updating it, to avoid lost updates in multi-user systems.

Multithreading
Running several threads, each a separate path of execution, within one process so they share its memory and code.

Modulus 11
A check digit method in which each digit is multiplied by a weighting, the products are added, and the check digit is found from the remainder after dividing by 11 (0 gives 0, 1 gives X, otherwise 11 minus the remainder).

Control total
A batch total formed by adding up a meaningful field on every document, such as total hours worked, and comparing it with the computer total.

Batch processing
Processing in which transactions are collected over a period in a transaction file and processed together as one job at a later time, without user intervention.

Real-time processing
Processing in which each transaction or input is dealt with as soon as it occurs, so the master file is always up to date; real-time control systems also guarantee a response fast enough to affect the next input.

Multitasking
The apparent running of several programs at once by rapidly switching the processor between them.

Multi-user
A system that allows many users to share the same computer's resources at the same time, each with their own session and permissions.

C C# example

`Modulus 11 in C#`

```
string number = "3604";
int weight = number.Length + 1;
int total = 0;
foreach (char c in number)
{
    total += (c - '0') * weight;
    weight--;
}
int remainder = total % 11;
string check;
if (remainder == 0) check = "0";
else if (remainder == 1) check = "X";
else check = (11 - remainder).ToString();
Console.WriteLine(number + check);   // 36048
```

Exam focus

Check digit questions are usually Calculate questions worth three to four marks: show the weightings, the products, the checksum, the remainder and the final check digit, and remember the rules for remainders of 0 and 1. If asked to show an error is detected, recalculate including the check digit with weighting 1 and state that the remainder is not zero. Mark schemes for batch totals credit a clear definition of each type and an applied example using a field from the scenario: count of documents for a batch total, a meaningful sum for a control total and a meaningless sum, such as IDs, for a hash total. Explaining why a hash total catches errors a batch total misses earns extra credit. For processing methods, name the method and justify it from the scenario: how quickly results are needed, whether the master file must always be current, and the volume of data. A booking system or cash machine is real-time transaction (online) processing because the master file is updated at once; anti-lock brakes are real-time control. Explain record locking in terms of preventing double-booking or lost updates.

? Check yourself

1. State the purpose of a check digit. [1]
2. A library in Strabane issues borrower numbers that end in a Modulus 11 check digit. Work out the check digit for the base number 2731, showing your working. [3]
3. A library card number is stored as 1250X. Use the Modulus 11 method to show that this number has been entered incorrectly. [3]

Answers: 1. To detect errors when a code number is entered or scanned, e.g. transposition/transcription errors (1) 2. Weightings 5, 4, 3, 2; products 10, 28, 9, 2 (1) / checksum 49; 49 ÷ 11 = 4 remainder 5 (1) / check digit 11 − 5 = 6, full number 27316 (1) 3. Weightings 5, 4, 3, 2, 1 with X = 10 (1) / products 5 + 8 + 15 + 0 + 10 = 38 (1) / 38 ÷ 11 = 3 remainder 5; remainder is not 0, so the number is invalid (1). Alternative: check digit for 1250 is 28 MOD 11 = 6, 11 − 6 = 5, not X.